Physical Chemistry of Materials

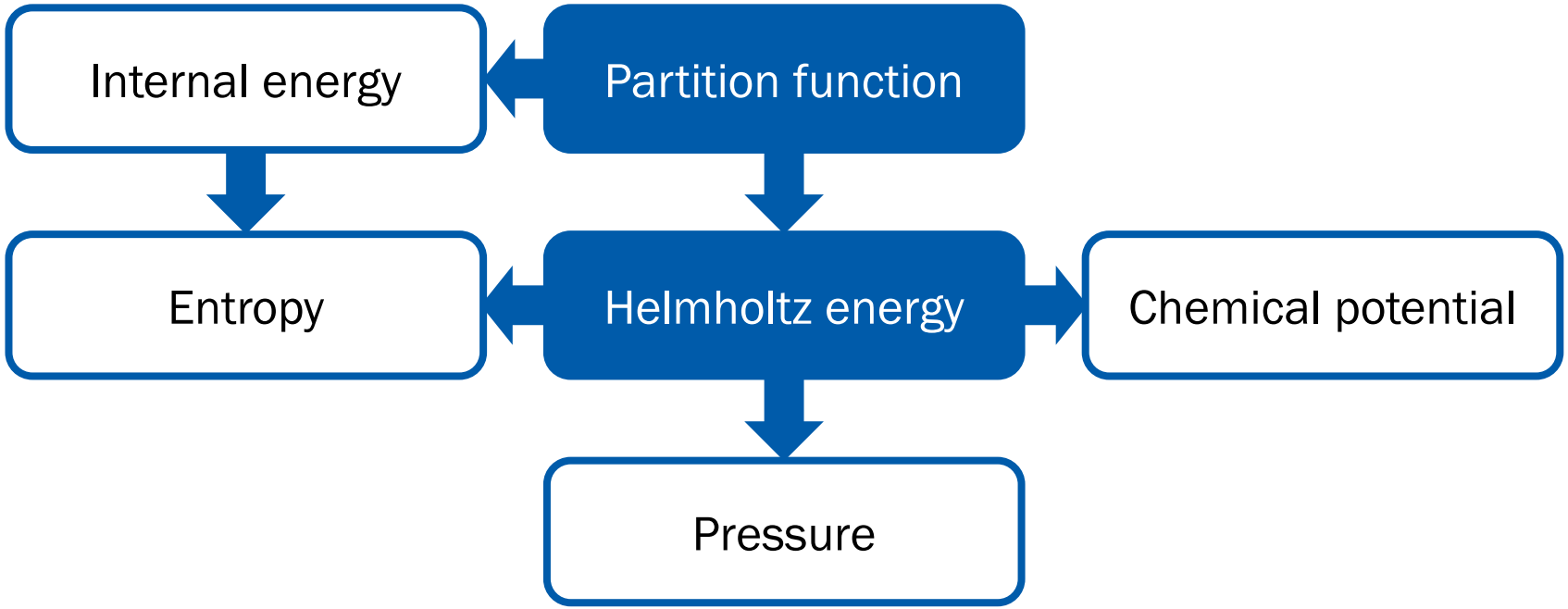
Lecture 06

Ideal Gas

Chemical Equilibrium

Last Class

Z and Thermodynamics



Last Class

Multi-Particle PFs

- Independent & distinguishable particles

$$Z = z^N \ (z\text{: one-particle partition function})$$

- Independent & indistinguishable particles

$$Z = \frac{1}{N!} z^N$$

- Interacting particles: later

Last Class

Classical Partition Function

$$z = \sum_s e^{-\beta E(s)} \quad \rightarrow \quad z = \frac{1}{\sigma_0} \int \int e^{-\beta E(x,p)} \, \mathrm{d}x \mathrm{d}p$$

for a 1-dimensional single particle

PF for 3D Ideal Gas

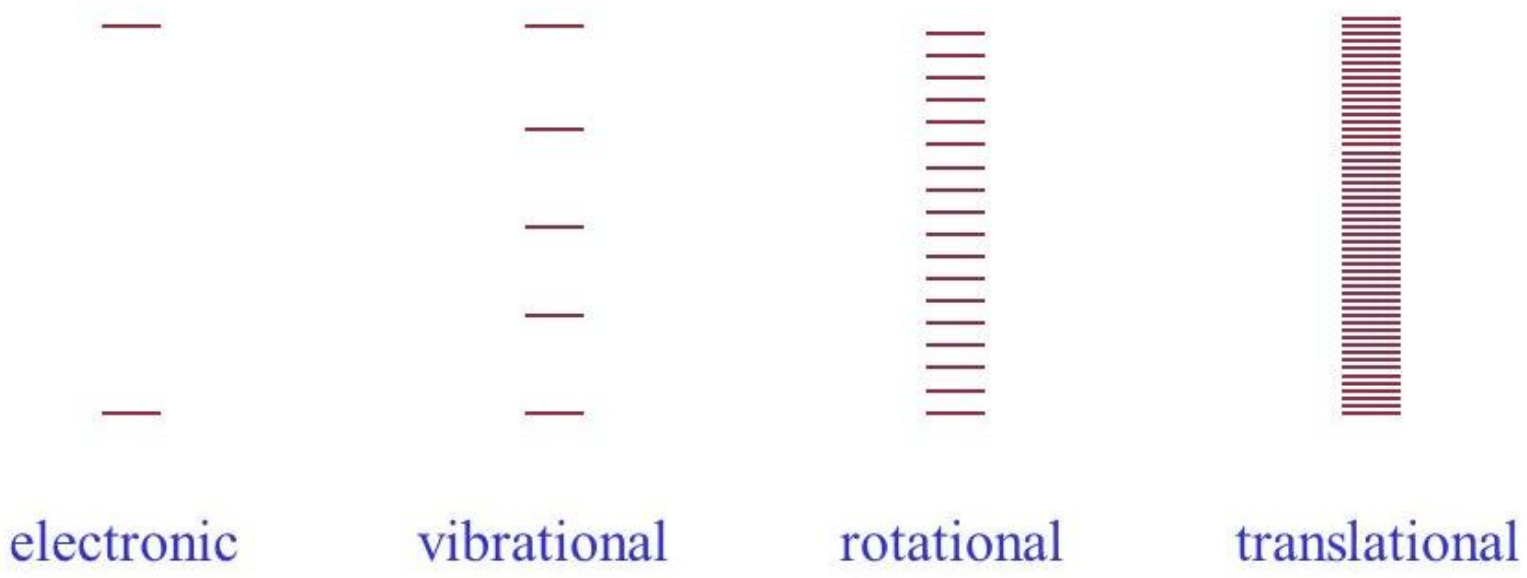
- Classical version

$$Z = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{\sigma_0^2}\right)^{3N/2}$$

- Quantum version

$$Z = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2}$$

Molecular Energies in QM



(Image: https://slideplayer.com/slide/6282680/)

QM Model Systems for Molecules

- Translation: particle in a box

$$E_n = \frac{h^2}{8ma^2} n^2$$

- Vibration: simple harmonic oscillator

$$E_v = h\nu\left(v + \frac{1}{2}\right)$$

- Rotation: rigid rotor

$$E_J = \hbar^2 J(J+1)/2I \text{ and } g_J = 2J + 1$$

Energy and Partition Function

- Monatomic molecules

$$E = E_{\text{trans}} + E_{\text{elec}}$$

$$z = z_{\text{trans}} z_{\text{elec}}$$

- Diatomic and polyatomic molecules

$$E = E_{\text{trans}} + E_{\text{vib}} + E_{\text{rot}} + E_{\text{elec}}$$

$$z = z_{\text{trans}} z_{\text{vib}} z_{\text{rot}} z_{\text{elec}}$$

Electronic Partition Function

At room temperature, almost all molecules are in their ground electronic states (no thermal excitation)

$$z_{\text{elec}} = \sum_{\substack{\text{electronic} \\ \text{state } s}} e^{-E_s/k_{\text{B}}T} \approx g_0 e^{-E_0/k_{\text{B}}T}$$

Translational Partition Function

- Particle in a box
- Already calculated in the previous section

$$z_{\text{trans}} = \left(\frac{2\pi m k_{\text{B}} T}{h^2}\right)^{3/2} V$$

- Consistent for classical and quantum treatments!

Vibrational Partition Function

- Simple harmonic oscillator
- Classical treatment

$$z_{\text{vib}} = \frac{1}{\sigma_0}\int\int e^{-\beta E(x,p)}\,\mathrm{d}x\mathrm{d}p = \frac{1}{h}\int\int e^{-\beta E(x,p)}\,\mathrm{d}x\mathrm{d}p$$

$$E(x,p) = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kx^2 + \frac{p^2}{2m}$$

$$z_{\text{vib}} = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\beta\left(\frac{1}{2}kx^2+\frac{p^2}{2m}\right)}\,\mathrm{d}x\mathrm{d}p$$

Classical 1-Dimensional Spring

$$z_{\text{vib}} = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\beta\left(\frac{1}{2}kx^2+\frac{p^2}{2m}\right)}\,\mathrm{d}x\mathrm{d}p$$

$$z_{\text{vib}} = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{\beta kx^2}{2}-\frac{\beta p^2}{2m}}\,\mathrm{d}x\mathrm{d}p$$

$$z_{\text{vib}} = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{\beta kx^2}{2}}\times e^{-\frac{\beta p^2}{2m}}\,\mathrm{d}x\mathrm{d}p$$

$$z_{\text{vib}} = \frac{1}{h}\left[\int_{-\infty}^{\infty} e^{-\frac{\beta kx^2}{2}}\,\mathrm{d}x\right]\left[\int_{-\infty}^{\infty} e^{-\frac{\beta p^2}{2m}}\,\mathrm{d}p\right]$$

Gaussian Integrals

$$z_{\text{vib}} = \frac{1}{h}\left[\int_{-\infty}^{\infty} e^{-\frac{\beta k x^2}{2}}\,\mathrm{d}x\right]\left[\int_{-\infty}^{\infty} e^{-\frac{\beta p^2}{2m}}\,\mathrm{d}p\right]$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2}\,\mathrm{d}x = \sqrt{\pi/\alpha}$,

$$z_{\text{vib}} = \frac{1}{h}\sqrt{\frac{2\pi}{\beta k}}\sqrt{\frac{2m\pi}{\beta}} = \frac{2\pi}{\beta h}\sqrt{\frac{m}{k}}$$

Vibrational Partition Function

- Simple harmonic oscillator
- Classical treatment

$$z_{\text{vib}} = \frac{2\pi}{\beta h}\sqrt{\frac{m}{k}} = \frac{k_{\text{B}}T}{h\nu}$$

- Quantum treatment

$$z_{\text{vib}} = \sum_{v=0}^{\infty} e^{-\beta E_v} = \sum_{v=0}^{\infty} e^{-\beta h\nu\left(v+\frac{1}{2}\right)}$$

Quantum Version of z_{vib}

$$z_{\mathrm{vib}} = \sum_{v=0}^{\infty} e^{-\beta h\nu\left(v+\frac{1}{2}\right)}$$

Let $x = e^{-\beta h\nu}$; for $x < 1$,

$$z_{\mathrm{vib}} = x^{1/2} \sum_{v=0}^{\infty} x^{v} = \frac{x^{1/2}}{1-x}$$

$$\therefore \; z_{\mathrm{vib}} = \frac{e^{-\beta h\nu/2}}{1-e^{-\beta h\nu}}$$

Comparison of Two Versions

- Classical treatment

$$z_{\mathrm{vib,CM}} = \frac{k_{\mathrm{B}}T}{h\nu}$$

- Quantum treatment

$$z_{\mathrm{vib,QM}} = \frac{e^{-h\nu/2k_{\mathrm{B}}T}}{1 - e^{-h\nu/k_{\mathrm{B}}T}}$$

- For $T \to \infty$ (classical limit), using $e^{-x} \approx 1 - x$ for $x \ll 1$,

$$z_{\mathrm{vib,QM}} \approx \frac{k_{\mathrm{B}}T}{h\nu}$$

Rotational Partition Function

- Rigid rotor
- Classical treatment (requires messy math)

$$z_{\text{rot}} = \frac{8\pi^2 I}{\beta h^2}$$

- Quantum treatment

$$z_{\text{rot}} = \sum_{J=0}^{\infty} g_J e^{-\beta E_J} = \sum_{J=0}^{\infty} (2J+1) e^{-\beta \hbar^2 J(J+1)/2I}$$

Quantum Version of z_{rot}

$$z_{\mathrm{rot}} = \sum_{J=0}^{\infty} (2J+1) e^{-\beta \hbar^2 J(J+1)/2I}$$

Take a continuum approximation (at high T):

$$z_{\mathrm{rot}} = \int_0^{\infty} (2J+1) e^{-\beta \hbar^2 J(J+1)/2I} \, \mathrm{d}J$$

Letting $x = J(J+1)$,

$$z_{\mathrm{rot}} = \int_0^{\infty} e^{-\beta \hbar^2 x/2I} \, \mathrm{d}x = \frac{2I}{\beta \hbar^2}$$

Rotational Partition Function

- Rigid rotor
- Classical treatment (requires messy math)

$$z_{\text{rot}} = \frac{8\pi^2 I}{\beta h^2}$$

- Quantum treatment

$$z_{\text{rot}} = \frac{2I}{\beta \hbar^2} = \frac{8\pi^2 I}{\beta h^2}$$

Consistent!

Symmetry Number

- Homonuclear diatomic molecule: 180° rotation leads to the same configuration

  → Correction required!

- Symmetry number σ

$$z_{\mathrm{rot}} = \frac{2Ik_{\mathrm{B}}T}{\sigma\hbar^2}$$

"Number of indistinguishable orientations of the molecule"

Linear Polyatomic Molecules

- Energy

$$E = E_{\text{trans}} + E_{\text{rot}} + \sum_j E_{\text{vib},j} + E_{\text{elec}}$$

- Partition functions

$$z = \underbrace{\left(\frac{2\pi M k_{\text{B}} T}{h^2}\right)^{3/2} V}_{z_{\text{trans}}} \cdot \underbrace{\frac{2 I k_{\text{B}} T}{\sigma \hbar^2}}_{z_{\text{rot}}} \cdot \underbrace{\prod_j \frac{e^{-h\nu_j/2k_{\text{B}}T}}{1 - e^{-h\nu_j/k_{\text{B}}T}}}_{z_{\text{vib}}} \cdot \underbrace{g_0 e^{-E_0/k_{\text{B}}T}}_{z_{\text{elec}}}$$

$$Z = \frac{1}{N!} z^N = \frac{1}{N!} \left(z_{\text{trans}} z_{\text{rot}} z_{\text{vib}} z_{\text{elec}}\right)^N$$

Linear Polyatomic Molecules

$$Z = \frac{1}{N!}\left(z_{\text{trans}} z_{\text{rot}} z_{\text{vib}} z_{\text{elec}}\right)^N$$

- Internal energy

$$U = -\frac{\partial \ln Z}{\partial \beta}$$

$$U = \frac{5}{2} N k_{\text{B}} T + \sum_j \left(\frac{N h \nu_j}{2} + \frac{N h \nu_j}{e^{h\nu_j / k_{\text{B}} T} - 1} \right) + N E_0$$

Law of Mass Action

For a chemical equilibrium

$$a\mathrm{A} + b\mathrm{B} \rightleftarrows c\mathrm{C} + d\mathrm{D},$$

the equilibrium constant is constant at fixed temperature

$$K_c = \frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}$$

Why?

Extent of Reaction

$$a\mathrm{A} + b\mathrm{B} \rightleftarrows c\mathrm{C} + d\mathrm{D}$$

Define ξ such that

$$\mathrm{d}n_\mathrm{A} = -a\ \mathrm{d}\xi$$

$$\mathrm{d}n_\mathrm{B} = -b\ \mathrm{d}\xi$$

$$\mathrm{d}n_\mathrm{C} = +c\ \mathrm{d}\xi$$

$$\mathrm{d}n_\mathrm{D} = +d\ \mathrm{d}\xi$$

Helmholtz Energy

$$\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V + \sum_i \mu_i \mathrm{d}n_i$$

For fixed temperature and volume,

$$\mathrm{d}A = \sum_i \mu_i \mathrm{d}n_i = (-a\mu_\mathrm{A} - b\mu_\mathrm{B} + c\mu_\mathrm{C} + d\mu_\mathrm{D})\mathrm{d}\xi$$

At equilibrium, $\mathrm{d}A = 0$ and

$$-a\mu_\mathrm{A} - b\mu_\mathrm{B} + c\mu_\mathrm{C} + d\mu_\mathrm{D} = 0$$

Chemical Potential

$$\mu = \left(\frac{\partial A}{\partial n}\right)_{T,V} = -k_{\mathrm{B}}T\left(\frac{\partial \ln Z}{\partial n}\right)_{T,V} = -RT\left(\frac{\partial \ln Z}{\partial N}\right)_{T,V}$$

For N molecules, $Z = \frac{1}{N!}z^N$ and

$$\ln Z \approx N \ln z - N \ln N + N$$

Hence,

$$\mu = -RT \ln z + RT \ln N + RT - RT = -RT \ln\left(\frac{Z}{N}\right)$$

Partition Functions in Action

$$-a\mu_{\mathrm{A}} - b\mu_{\mathrm{B}} + c\mu_{\mathrm{C}} + d\mu_{\mathrm{D}} = 0$$

$$\mu_i = -RT \ln\left(\frac{z_i}{N_i}\right)$$

Substitution leads to

$$RT\left[a \ln\left(\frac{z_{\mathrm{A}}}{N_{\mathrm{A}}}\right) + b \ln\left(\frac{z_{\mathrm{B}}}{N_{\mathrm{B}}}\right) - c \ln\left(\frac{z_{\mathrm{C}}}{N_{\mathrm{C}}}\right) - d \ln\left(\frac{z_{\mathrm{D}}}{N_{\mathrm{D}}}\right)\right] = 0$$

$$\therefore \frac{N_{\mathrm{C}}^{c} N_{\mathrm{D}}^{d}}{N_{\mathrm{A}}^{a} N_{\mathrm{B}}^{b}} = \frac{z_{\mathrm{C}}^{c} z_{\mathrm{D}}^{d}}{z_{\mathrm{A}}^{a} z_{\mathrm{B}}^{b}}$$

Equilibrium Constant

$$\frac{N_{\mathrm{C}}^{c} N_{\mathrm{D}}^{d}}{N_{\mathrm{A}}^{a} N_{\mathrm{B}}^{b}} = \frac{z_{\mathrm{C}}^{c} z_{\mathrm{D}}^{d}}{z_{\mathrm{A}}^{a} z_{\mathrm{B}}^{b}}$$

$$\frac{(N_{\mathrm{C}}/V)^{c}(N_{\mathrm{D}}/V)^{d}}{(N_{\mathrm{A}}/V)^{a}(N_{\mathrm{B}}/V)^{b}} = \frac{(z_{\mathrm{C}}/V)^{c}(z_{\mathrm{D}}/V)^{d}}{(z_{\mathrm{A}}/V)^{a}(z_{\mathrm{B}}/V)^{b}}$$

The left-hand side is the equilibrium constant:

$$K_c = \frac{[\mathrm{C}]^{c}[\mathrm{D}]^{d}}{[\mathrm{A}]^{a}[\mathrm{B}]^{b}} = \frac{(z_{\mathrm{C}}/V)^{c}(z_{\mathrm{D}}/V)^{d}}{(z_{\mathrm{A}}/V)^{a}(z_{\mathrm{B}}/V)^{b}}$$

K from Partition Function

$$H_2 + D_2 \rightleftarrows 2HD$$

$$K_c = \frac{[\mathrm{HD}]^2}{[\mathrm{H_2}][\mathrm{D_2}]} = \frac{(z_{\mathrm{HD}}/V)^2}{\left(z_{\mathrm{H_2}}/V\right)\left(z_{\mathrm{D_2}}/V\right)} = \frac{z_{\mathrm{HD}}^2}{z_{\mathrm{H_2}} z_{\mathrm{D_2}}}$$

For a diatomic molecule, recall that

$$z = z_{\mathrm{trans}} z_{\mathrm{rot}} z_{\mathrm{vib}} z_{\mathrm{elec}}$$

$$= \left(\frac{2\pi M k_{\mathrm{B}} T}{h^2}\right)^{3/2} V \cdot \frac{2I k_{\mathrm{B}} T}{\sigma \hbar^2} \cdot \frac{e^{-h\nu/2k_{\mathrm{B}}T}}{1 - e^{-h\nu/k_{\mathrm{B}}T}} \cdot g_0 e^{-E_0/k_{\mathrm{B}}T}$$

≈ 1 (at RT)

Equilibrium Constant

Substitution leads to

$$K_c = \left(\frac{M_{\mathrm{HD}}^2}{M_{\mathrm{H_2}} M_{\mathrm{D_2}}}\right)^{3/2} \frac{I_{\mathrm{HD}}^2}{I_{\mathrm{H_2}} I_{\mathrm{D_2}}} \frac{\sigma_{\mathrm{H_2}} \sigma_{\mathrm{D_2}}}{\sigma_{\mathrm{HD}}^2} e^{-\beta h\left(2\nu_{\mathrm{HD}} - \nu_{\mathrm{H_2}} - \nu_{\mathrm{D_2}}\right)/2}$$

Using $M_{\mathrm{AB}} = m_{\mathrm{A}} + m_{\mathrm{B}}$ and $I_{\mathrm{AB}} = \frac{m_{\mathrm{A}} m_{\mathrm{B}}}{m_{\mathrm{A}} + m_{\mathrm{B}}} R_{\mathrm{AB}}^2$,

$$K_c = \left(\frac{9}{8}\right)^{3/2} \times \frac{8}{9} \times 4 \times e^{-\beta h\left(2\nu_{\mathrm{HD}} - \nu_{\mathrm{H_2}} - \nu_{\mathrm{D_2}}\right)/2}$$

$$\therefore \; K_c = 4.24 \times e^{-(77.7\ \mathrm{K})/T}$$

Theory vs Experiment

T (K)	K_c (theory)	K_c (exp)
195	2.84	2.92
273	3.18	3.24
298	3.26	3.28
383	3.46	3.50
543	3.67	3.85